DATA COMMUNICATION AND NETWORKING	
Course Code: AD42	**Credits: 3:0:1**
Pre – requisites: Basic Concepts of Electrical and Electronics	**Contact Hours: 42L+14P**
Course Coordinator: Dr. Siddesh G M	

Course Content

Unit I

Data communication Fundamentals: Introduction, components, Data Representation, Data Flow; Networks – Network criteria, Physical Structures, Network Models, Categories of networks; Protocols, Standards, Standards organization. **Network Models** -Layered tasks; The OSI model – Layered architecture; Layers in the OSI model; TCP/IP Protocol suite; Addressing.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links:
  https://nptel.ac.in/courses/106105082
- Lab Component / Practical Topics : Wireshark Exercise on HTTP
- Impartus Recording: https://a.impartus.com/ilc/#/course/96149/452

Unit II

Digital Transmission Fundamentals: Analog & Digital data, Analog & Digital signals (basics); Transmission Impairment – Attenuation, Distortion and Noise; Data rate limits – Nyquist Bit Rate, Shannon Capacity. **Digital Transmission**: Digital-to-Digital conversion - Line coding, Line coding schemes (unipolar, polar, bipolar); Analog-to-Digital conversion - PCM.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: https://nptel.ac.in/courses/106105082
- Lab Component / Practical Topics : Socket Programming
- Impartus Recording: https://a.impartus.com/ilc/#/course/96149/452

Unit III

Error detection & correction: Cyclic codes – CRC, Polynomials, Checksum.
Network layer: Logical addressing - IPV4 addresses, Address space, notations, classful and classless addressing with problem solving, NAT, IPV6 addresses; IPV6

addresses; Network layer: Internet protocol - IPV4 datagram, fragmentation, checksum and options; IPV6 packet format, advantages and extension headers; Transition from IPV4 to IPV6.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: https://onlinecourses.nptel.ac.in/noc22_ee61/preview
- Lab Component / Practical Topics : Wireshark Exercise on IP, ICMP, DHCP
- Impartus Recording: https://a.impartus.com/ilc/#/course/96149/452

Unit IV

Network layer: Delivery, Forwarding, & Routing –Forwarding Techniques, Forwarding Process, Routing Table; Unicast routing protocols – Optimization, Intra and Inter domain routing, distance vector routing, link state routing, path vector routing.

Transport Layer - Process-to-Process delivery, User Datagram Protocol, Transmission Control Protocol.

- Pedagogy /Course delivery tools: Chalk and talk, Power point presentation, Videos
- Link : https://onlinecourses.nptel.ac.in/noc22_ee61/preview
- Lab Component / Practical Topics : Error detection and correction using CRC and Hamming Code
- Impartus Recording: https://a.impartus.com/ilc/#/course/96149/452

Unit V

Congestion control - Congestion, Congestion control, congestion control in TCP.

Application Layer: Domain Name System - Namespace, Domain name space, Distribution of Name space, DNS in internet, Resolution; Remote logging – TELNET; Electronic mail – Architecture, User Agent, Message Transfer Agent: SMTP; File transfer - File transfer protocol (FTP);

- Pedagogy /Course delivery tools: Chalk and talk, Power point presentation, Videos
- Link : https://onlinecourses.nptel.ac.in/noc22_ee61/preview
- Lab Component / Practical Topics : Multiplexing concepts
- Impartus Recording: https://a.impartus.com/ilc/#/course/96149/452

Lab Content

Sl. No.	Topics Covered
Cycle I	
1.	Wireshark tool demo.
2.	Trace Hypertext Transfer Protocol using packet sniffer and packet analyser.
3.	Trace Domain Name Server using packet sniffer and packet analyser.
4.	Trace File Transfer Protocol using packet sniffer and packet analyser.
5.	Trace Internet Control Message Protocol using packet sniffer and packet analyser
6.	Trace Dynamic Host Configuration Protocol using packet sniffer and packet analyser.
7.	Write a program for error detection using CRC-CCITT(16-bits).
8.	Write a program for error detection and correction using hamming code.
9.	Write a program to find the shortest path between vertices using bellman-ford algorithm.
10.	Write a program to find the shortest path between vertices using link Dijsktra algorithm.
11.	Write a program for congestion control using leaky bucket algorithm.
12.	Using TCP/IP sockets, write a client – server program where the client send the file name and the server send back the contents of the requested file if present.
13.	Design an RPC (Remote Procedure Call) application for a server performing integer arithmetic operations. Describe call and reply message contents, error handling, and client-server interaction.
14.	Design and implementation of a peer-to-peer application for performing basic Addition and displaying the contents of files shared among peers.

Note: Each Lab Session is of two hours duration/week

Suggested Learning Resources

Text Books:

1. Behrouz A. Forouzan, Data Communications and Networking, Fifth Edition, Tata McGraw-Hill Education, 2013.

Reference Books:

1. Introduction to Data Communications and Networking – Wayne Tomasi, Pearson Education, 2005.
2. Larry L. Peterson and Bruce S Davie: Computer Networks: A Systems Approach, Fifth Edition, Elsevier, 2011.
3. Tanenbaum: Computer Networks, 4th Ed, Pearson Education/PHI, 2003.
4. James F. Kurose and Keith W. Ross: Computer Networking: A Top- Down Approach, 6th edition, Addison-Wesley, 2013.

Course Outcomes (COs):

At the end of the course, the students will be able to:

1. Distinguish different communication models / protocol stacks (OSI & TCP/IP) and analyze the usage of appropriate network topology for a given scenario. (PO-1, 2, 3) (PSO-1,2).
2. Handle the issues associated with digital data signals and solve the problems on data transmission by measuring the performance parameters. (PO-1, 2, 3) (PSO-1, 2).
3. Apply different error detection, error correction as well as flow control strategies to solve error and solve the problems associated with transition from IPV4 to IPV6. (PO-1, 2, 3) (PSO-1, 2).
4. Use different protocols to achieve Address mapping, Error reporting & routing. (PO-1, 2, 3, 5) (PSO-2).
5. Describe the working of various application layer services. (PO-2, 3) (PSO-2).

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	**Marks**	**e Outcomes (COs) addressed**
Internal Test-I (CIE-I)	30	CO1, CO2
Internal Test-II CIE-II)	30	CO3, CO4, CO5
Average of the two CIE shall be taken for 30 marks		
Other Components		
Lab Test	10	CO1, CO2, CO3, CO4, CO5
Tool Demonstration	10	CO1, CO2, CO3, CO4, CO5
The Final CIE out of 50 Marks = Average of two CIE tests for 30 Marks+ Marks scored in Lab Test +Marks scored in Demo		
Semester End Examination (SEE)		
Course End Examination (Answer One full question from each Unit- Internal Choice)	100	CO1, CO2, CO3, CO4, CO5